Adithya G S

Full Stack Engineer | Java & Spring Boot

Bengaluru, India | +91 8073790117 | adithyaaadi563@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Full Stack Engineer with **3+ years** building and operating production applications, specializing in Java, Spring Boot, backend architecture, distributed systems, and cloud engineering. Experienced in reliable data pipelines, secure integrations, asynchronous processing, and hands-on ownership from technical design through deployment and production support.

Technical Skills

Languages – Java, TypeScript, JavaScript, SQL, Python

Backend & APIs – Spring Boot, Spring WebFlux, Spring Security, Spring Data JPA, Hibernate, Node.js, Express, OAuth2, JWT, RBAC

Databases & Search – PostgreSQL, TimescaleDB, MongoDB, Redis, Elasticsearch, pgvector

Messaging & Queues – Kafka, AWS SQS, BullMQ

Cloud & DevOps – AWS (Lambda, S3, RDS, ECS, ECR, Lightsail), Azure Functions, Docker, Kubernetes, GitHub Actions

AI Engineering – LLM Tool Calling, AI Agents, MCP, LangChain, LangGraph, Embeddings, Semantic Retrieval, Clustering

Frontend – React, Next.js, Redux, Redux-Saga, TanStack Query, Material UI, shadcn/ui

Testing & Observability – JUnit, Mockito, Jest, Grafana, Prometheus, CloudWatch

Experience

Nighthack Labs **Jun 2024 – Present**

Software Engineer L3 | Bengaluru, India

- Led an analytics redesign after identifying **N+1 queries**, expensive joins, missing indexes, and reporting contention. Evaluated storage alternatives, migrated **10M+ records** with validation and recovery planning, and introduced indexed reporting models, reducing API latency from **12–40 seconds to under 2 seconds**.
- Re-architected advertising data ingestion into **prioritized, dependency-aware ELT workflows** with incremental reporting transformations, enabling earlier access to critical data and reducing time to the first usable dashboard from weeks to **1–2 hours**.
- Implemented **OAuth2 account connections** across advertising, analytics, and CRM platforms, handling authorization callbacks, encrypting provider tokens, and enforcing workspace-scoped access for **100+ connected accounts**.
- Redesigned synchronous competitor-ad retrieval into **fault-tolerant asynchronous workflows**, reducing API timeouts and failures during large scraping jobs. Introduced **provider-agnostic execution**, persistent job dispatch, concurrency-safe claiming, idempotent processing, backoff retries, and **SSE-based progress updates**.
- Replaced resource-contending workers with independently scalable **SQS and Lambda** workflows, isolating workloads and tracking job state while reducing **p95 background-job completion latency by around 50–60%**.
- Redesigned memory-intensive video exports into asynchronous jobs supporting **roughly 100 videos per export and up to 15 GB** through bounded parallel downloads, ZIP creation, **S3 multipart uploads**, and progress and completion notifications via **SSE**.
- Built **pgvector-backed semantic retrieval** with similarity search and clustering to supply representative advertising themes for LLM insights, reducing LLM input volume and enabling faster analysis with richer context.
- Built backend services for a **multi-tenant low-code SaaS platform**, implementing secure, workspace-scoped integration token management and reactive usage metering. Batched usage events in Elasticsearch and integrated **Lago and Stripe** billing with reconciliation for consumption-based charging.
- Owned architecture, design documentation, testing, CI/CD, and production support. **Led a two-engineer backend team**, coordinating task allocation, PR reviews, and deployments while contributing hands-on to development.

Incture Technologies **Aug 2023 – May 2024**

Associate Software Engineer | Bengaluru, India

- Developed and tested enterprise accounts-payable workflows and REST APIs using **Java, Spring Boot, Hibernate, and SAP HANA**, implementing business validation, persistence, and exception handling.
- Collaborated with client stakeholders to gather requirements and built a **proof of concept** for translating Russian-language invoices into English to support invoice processing.

Education

PES College of Engineering, Mandya **2019 – 2023**

B.E., Electronics and Communication Engineering | CGPA 9.02/10